

# Using Python on cRIO

(!! To Enable Vibe Coding !!)



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Printing variables in Python

Reading analog input with NI DAQ...

Create code snippets

D

Dan

Pro plan

Printing variables in Python

want to resume a previously running FPGA.

- Context manager ( `with` ) — always use `with nifpga.Session(...)` so the session is properly closed and the FPGA is left in a clean state even if an exception occurs.
- The NI-RIO driver must be installed on your system in addition to the Python package.

ALL PYTHON CODE WAS VIBED...

DON'T TELL MY BOSS...

Great. Now make it so this application reads the U64 indicator continuously

python

```
import nifpga
import time
import signal
import sys

def run_fpga_continuous(
    bitfile_path,
    resource,
    indicator_name,
    control_name,
    control_value=True,
    poll_interval=0.1
):
    """
```

Reply...

Sonnet 4.6



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CLA, CLED, 20 years “doing” w/ NI

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We are here to help!

- Pull us in to design conversations
- Let us help de-risk projects
- Tell us what you want to learn about next

# Assumptions

- cRIO is already:
  - Imaged
  - Has NI DAQmx drivers installed
  - Has NI RIO drivers installed
  - User name and password configured

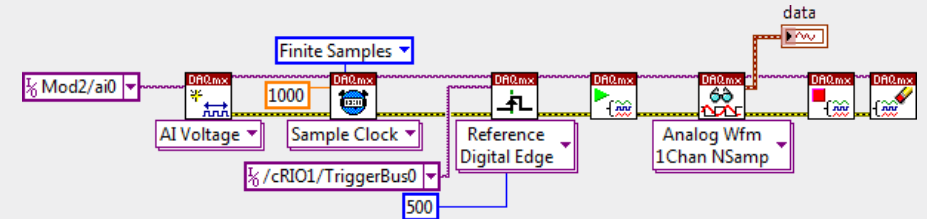
(Start here if you need help: <https://www.ni.com/docs/en-US/bundle/crio-904x/page/getting-started-overview.html>)

- Your Laptop (Optional but helpful)
  - Has IDE & Python installed
  - Has NI DAQmx installed
  - Has LabVIEW + RT + FPGA installed (IF wanting to program FPGA directly)

# Pick the IO API (per slot)

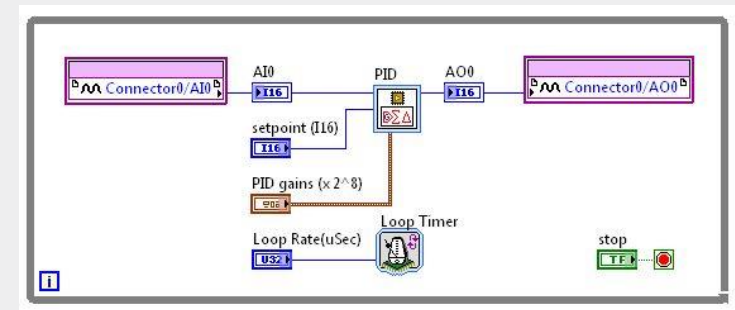
## Real-Time (NI-DAQmx)

- Simple API for waveform and control from processor
- Supports control loop rates up to 5 kHz
- All channels in a task are synchronized
- Real-Time uses appropriate driver, such as NI-DAQmx or XNET



## LabVIEW FPGA

- Single-point data acquisition and control from LabVIEW FPGA
- Supports control loop rates up to 10s of MHz
- Hardware-level determinism and reliability for control loops
- In-line signal processing for live filtering, calculations, and more

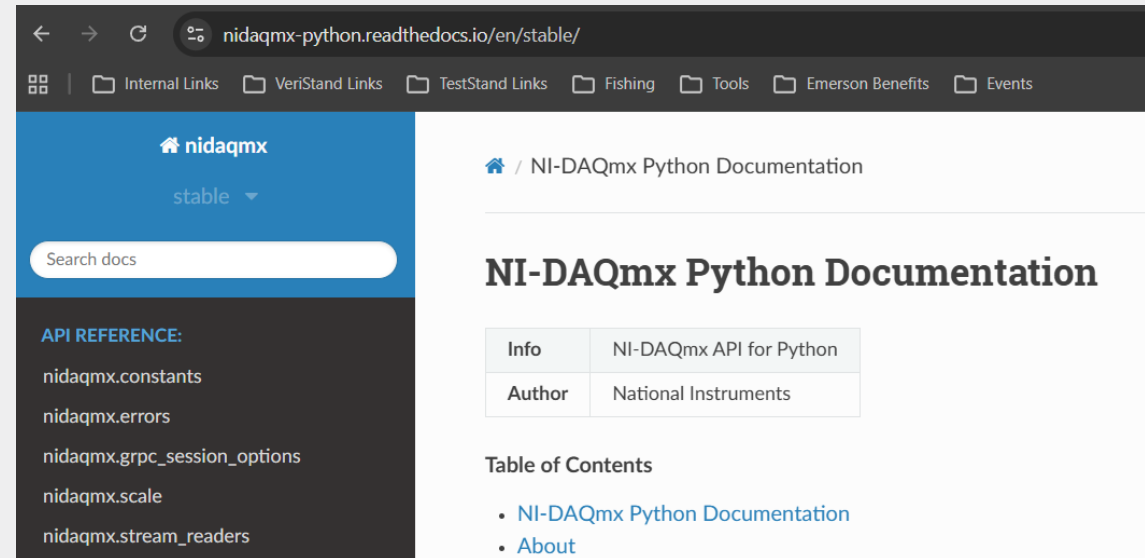


# Python APIs for NI DAQmx & FPGA

## Documentation = Vibe-able

DAQmx Documentation: <https://nidaqmx-python.readthedocs.io/en/stable/>

FPGA Documentation: <https://nifpga-python.readthedocs.io/en/latest/>



The screenshot shows a web browser displaying the NI-DAQmx Python Documentation page. The browser's address bar shows the URL `nidaqmx-python.readthedocs.io/en/stable/`. The page has a dark blue sidebar on the left with the `nidaqmx` logo and a 'stable' version selector. Below the logo is a search bar labeled 'Search docs'. Underneath the search bar, under the heading 'API REFERENCE:', is a list of links: `nidaqmx.constants`, `nidaqmx.errors`, `nidaqmx.grpc_session_options`, `nidaqmx.scale`, and `nidaqmx.stream_readers`. The main content area has a white background and a dark blue header with the text '/ NI-DAQmx Python Documentation'. Below the header, the title 'NI-DAQmx Python Documentation' is displayed in a large, bold font. Underneath the title is a table with two rows: 'Info' with the value 'NI-DAQmx API for Python' and 'Author' with the value 'National Instruments'. Below the table, the heading 'Table of Contents' is followed by a list of links: 'NI-DAQmx Python Documentation' and 'About'.

|        |                         |
|--------|-------------------------|
| Info   | NI-DAQmx API for Python |
| Author | National Instruments    |

Table of Contents

- [NI-DAQmx Python Documentation](#)
- [About](#)

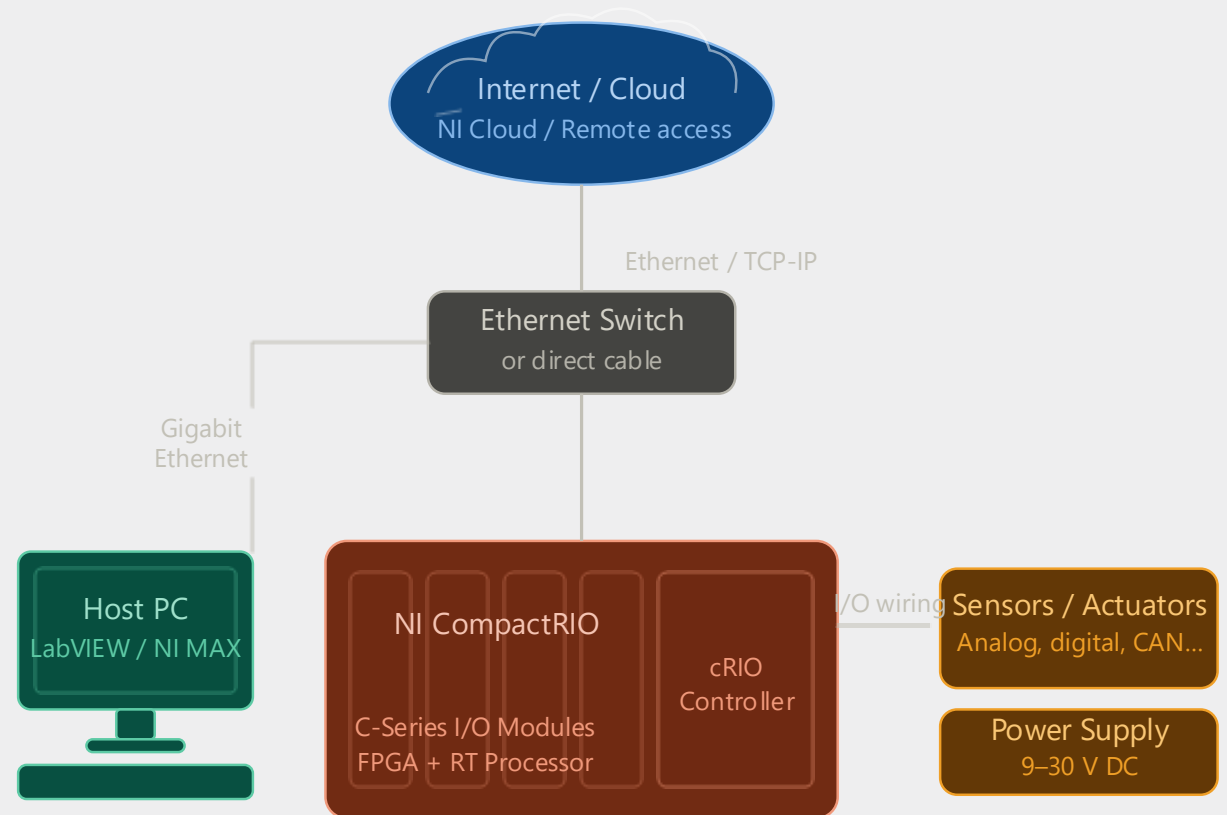
# Connect cRIO to Internet & Laptop

Connect your laptop to a switch

Connect your cRIO to a switch

Make sure they are all on the same subnet (255.255.255.0)

Connect the switch to the internet (router, etc)



# Install Base Image and Programming Environment

Pick your Base Image

Pick your Programming Environment

Real-Time Software - NI-cRIO-9047 (192.168.68.57)

|                                                                                                                                             |                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Base System Image</b><br>Linux RT System Image 2026 Q3 |  <b>Programming Environment</b><br>Other (Select to use C, C++, Python, etc.) |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

Real-Time Software - NI-cRIO-9047 (192.168.68.57)

**Select the programming environment.** [Learn more](#)

☐ LabVIEW

Version  
2026

☐ LabVIEW NXG

☒ Other (Select to use C, C++, Python, etc.)



# Install Required Drivers

Install NI DAQmx on Host

Install NI DAQmx on cRIO

Make sure the versions match

The screenshot displays the NI Package Manager interface. On the left, a tree view shows the system hierarchy under 'My System' > 'Remote Systems' > 'NI-cRIO-9045-01CEEDF1'. The 'Software' section is expanded, listing various components including 'NI-DAQmx 26.0.0.49428-0+f276'. On the right, the 'INSTALLED' tab is active, showing a search for 'DAQmx' with 1 result. The result table lists the installed driver.

| Name        | Value                                                                     |
|-------------|---------------------------------------------------------------------------|
| Description | Provides support for NI data acquisition and signal conditioning devices. |
| Version     | 26.0.0.49428-0+f276                                                       |

| Category                          | Maintainer           | Clear Filters | Products only | Search |
|-----------------------------------|----------------------|---------------|---------------|--------|
| 1 result                          |                      |               |               |        |
| Name                              | Maintainer           | Category      | Version       |        |
| <input type="checkbox"/> NI-DAQmx | National Instruments | Drivers       | 2026 Q1       |        |

# Set Modules to use NI-DAQmx

In Max, under cRIO, per module, set

- Program Mode = NI-DAQmx

Note:

Per module means you can mix NI DAQmx & FPGA access

The screenshot displays the NI Measurement & Automation Explorer (MAX) interface. The title bar reads "4: NI 9205 'Mod4' - Measurement & Automation Explorer". The menu bar includes File, Edit, View, Tools, and Help. The left pane shows a tree view of the system hierarchy: My System, Remote Systems, NI-cRIO-9045-01CEEDF1, Data Neighborhood, Devices and Interfaces, NI cRIO-9045 "RIO0", ASRL1::INSTR, ASRL2::INSTR, NI cRIO-9045 "cRIO1", and a list of modules. The module "4: NI 9205 'Mod4'" is selected and highlighted. The right pane shows the "Settings" tab for this module, with the following fields:

| Settings                   |                      |
|----------------------------|----------------------|
| Name                       | Mod4                 |
| Vendor                     | National Instruments |
| Model                      | NI 9205              |
| Serial Number              | 0192A01C             |
| Hardware Revision          | J                    |
| Slot Number                | 4                    |
| Status                     | Present              |
| Program Mode               | Real-Time (NI-DAQmx) |
| Current Device Temperature | 30.9°C               |

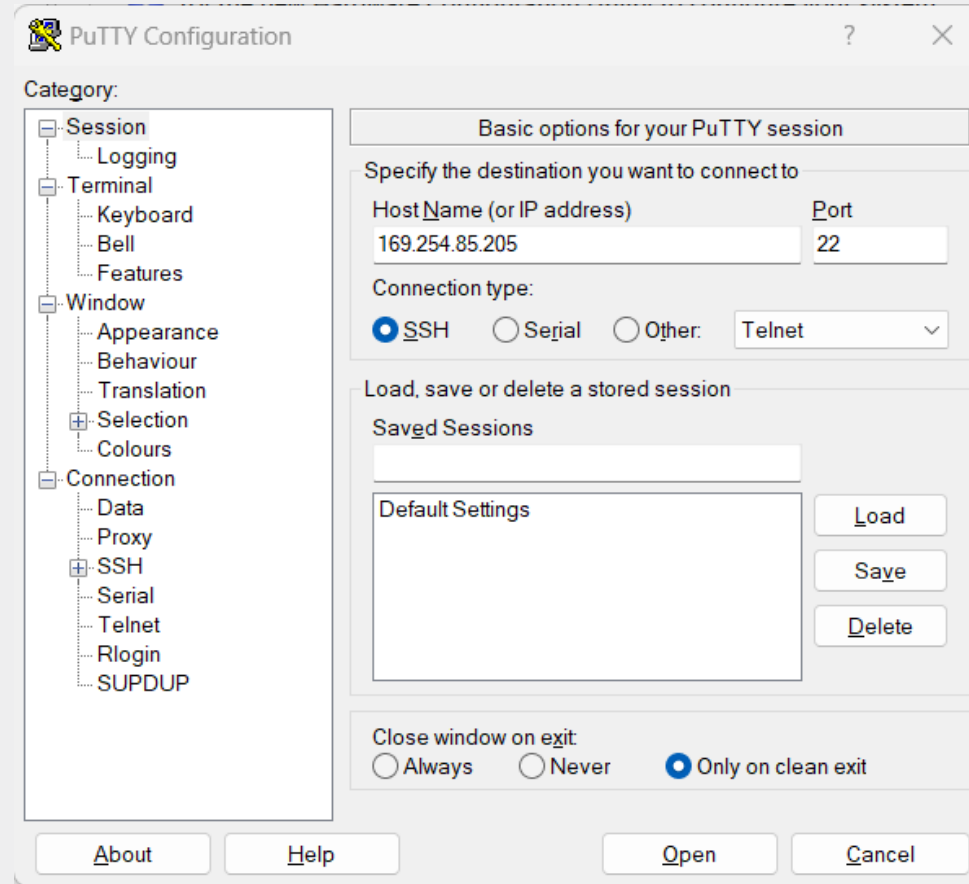
# Using PuTTY (SSH) with cRIO

## Connect

### Set

- Host Name to cRIO IP Address
- Port to 22
- Connection type to SSH

### Hit Open

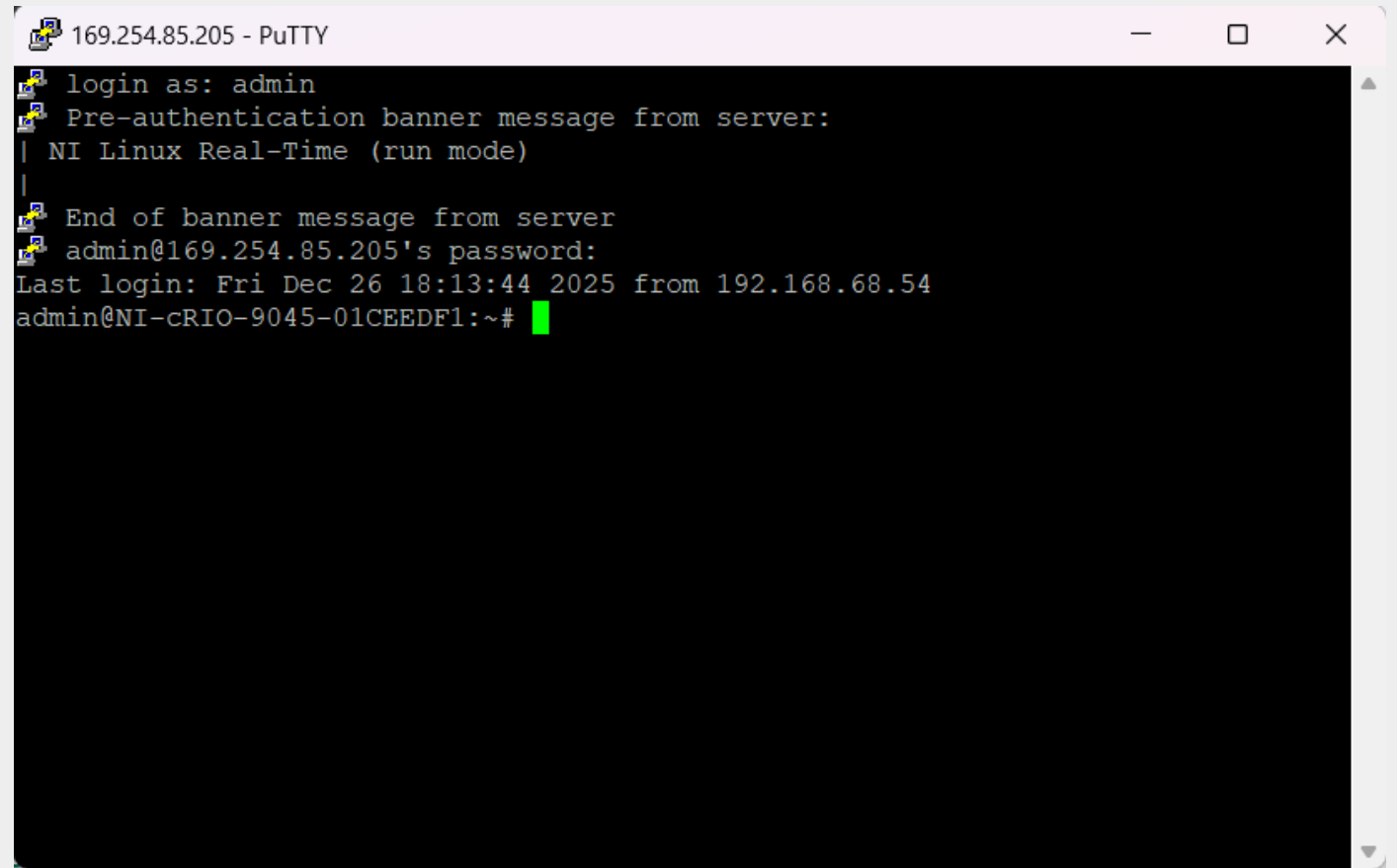


# Using PuTTY (SSH) with cRIO

## Log In

Enter User Name (& hit enter)

Enter Password (& hit enter)



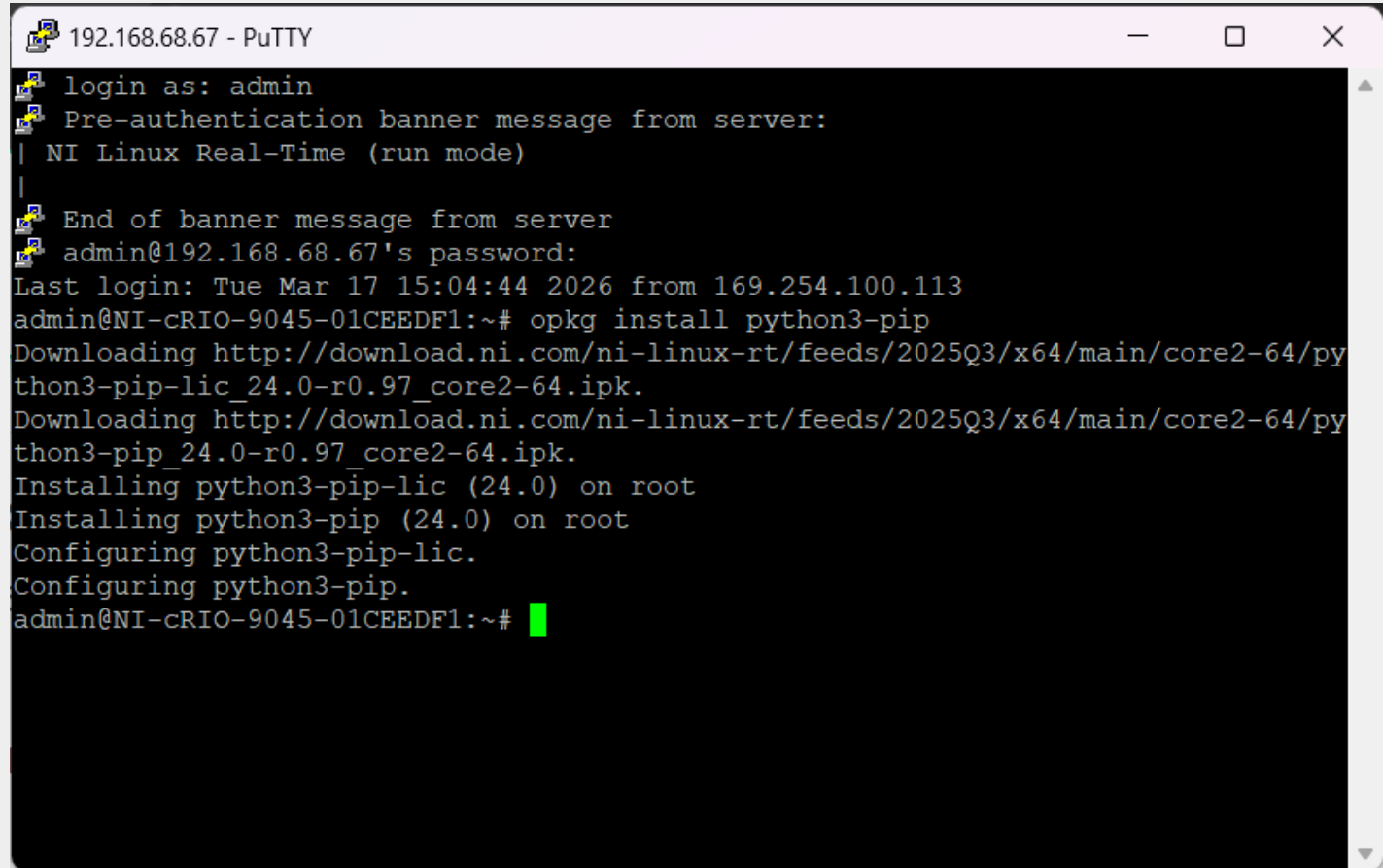
```
169.254.85.205 - PuTTY
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
admin@169.254.85.205's password:
Last login: Fri Dec 26 18:13:44 2025 from 192.168.68.54
admin@NI-cRIO-9045-01CEEDF1:~#
```

# Using PuTTY (SSH) with cRIO

## Install Python

In root, type:

- `opkg install python3-pip` (Installs Python)



```
192.168.68.67 - PuTTY
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
admin@192.168.68.67's password:
Last login: Tue Mar 17 15:04:44 2026 from 169.254.100.113
admin@NI-cRIO-9045-01CEEDF1:~# opkg install python3-pip
Downloading http://download.ni.com/ni-linux-rt/feeds/2025Q3/x64/main/core2-64/py
thon3-pip-lic_24.0-r0.97_core2-64.ipk.
Downloading http://download.ni.com/ni-linux-rt/feeds/2025Q3/x64/main/core2-64/py
thon3-pip_24.0-r0.97_core2-64.ipk.
Installing python3-pip-lic (24.0) on root
Installing python3-pip (24.0) on root
Configuring python3-pip-lic.
Configuring python3-pip.
admin@NI-cRIO-9045-01CEEDF1:~#
```

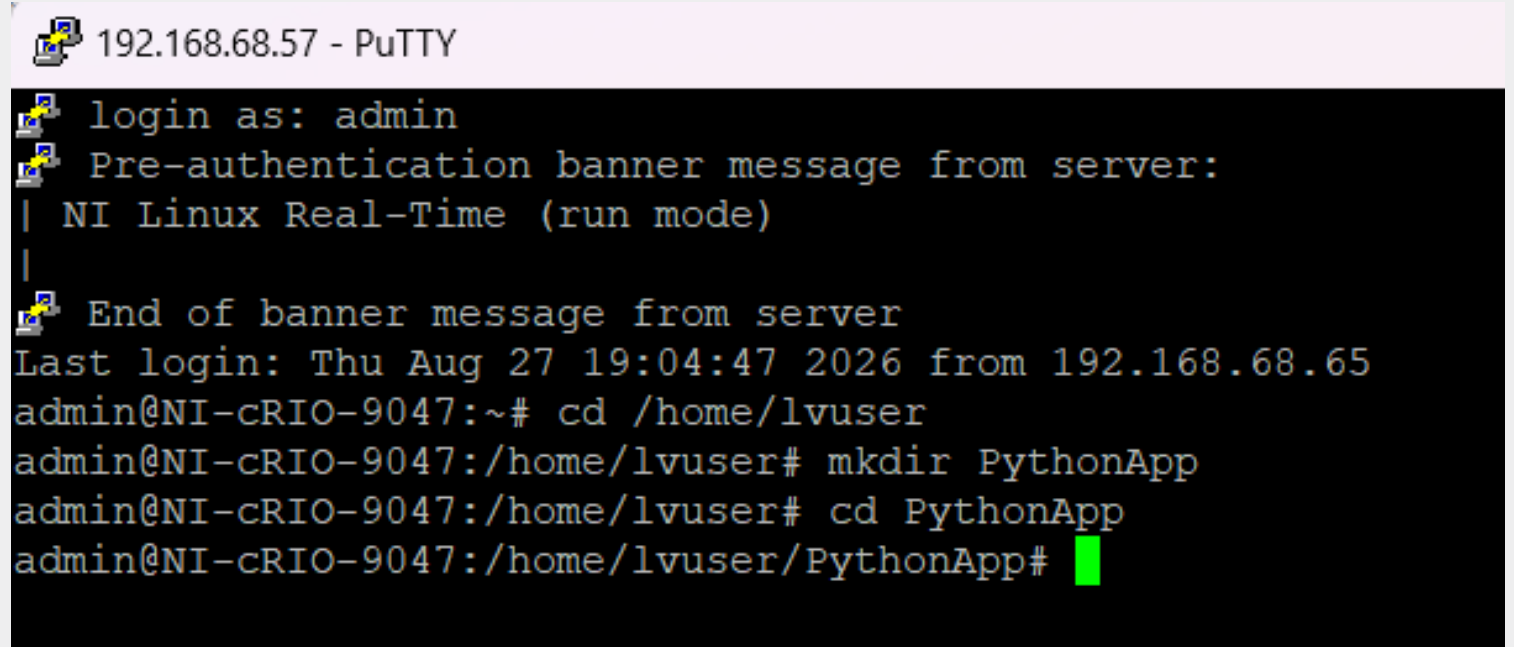
# Using PuTTY (SSH) with cRIO

## Create the directory

In root, type:

- `cd /home/lvuser`
- `mkdir PythonApp`
- `cd PythonApp`

Note: You should be in `/home/lvuser/PythonApp`



```
192.168.68.57 - PuTTY
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
Last login: Thu Aug 27 19:04:47 2026 from 192.168.68.65
admin@NI-cRIO-9047:~# cd /home/lvuser
admin@NI-cRIO-9047:/home/lvuser# mkdir PythonApp
admin@NI-cRIO-9047:/home/lvuser# cd PythonApp
admin@NI-cRIO-9047:/home/lvuser/PythonApp#
```

# Using PuTTY (SSH) with cRIO

## Create the Virtual Environment & Activate

In /home/lvuser/PythonApp, type:

- `python -m venv venv`
- `source venv/bin/activate`

```
admin@NI-cRIO-9047:/home/lvuser/PythonApp# python -m venv venv
admin@NI-cRIO-9047:/home/lvuser/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp#
```

Note: You should now see ((venv))

# Using PuTTY (SSH) with cRIO

## Upgrade PIP (Package Installer)

In ((venv)) /home/lvuser/PythonApp, type:

- `python -m pip install --upgrade pip`

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# python -m pip install --upgrade pip
```



# Using PuTTY (SSH) with cRIO

## Install NI DAQmx Package

In ((venv)) /home/lvuser/PythonApp, type:

- pip install nidaqmx

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# pip install nidaqmx
Collecting nidaqmx
  Using cached nidaqmx-1.6.0-py3-none-any.whl.metadata (17 kB)
Collecting click>=8.0.0 (from nidaqmx)
  Using cached click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting deprecation>=2.1 (from nidaqmx)
  Using cached deprecation-2.1.0-py2.py3-none-any.whl.metadata (4.6 kB)
Collecting distro>=1.9.0 (from nidaqmx)
  Using cached distro-1.9.0-py3-none-any.whl.metadata (6.8 kB)
Collecting hightime>=0.2.2 (from nidaqmx)
  Using cached hightime-1.0.0-py3-none-any.whl.metadata (2.7 kB)
Collecting nitypes>=1.0.1 (from nidaqmx)
  Using cached nitypes-1.1.0-py3-none-any.whl.metadata (7.4 kB)
Collecting numpy>=1.22 (from nidaqmx)
  Using cached numpy-2.5.2-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Collecting python-decouple>=3.8 (from nidaqmx)
  Using cached python_decouple-3.8-py3-none-any.whl.metadata (14 kB)
Collecting requests>=2.25.0 (from nidaqmx)
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
```

# Using PuTTY (SSH) with cRIO

## Install NI FPGA Package

In ((venv)) /home/lvuser/PythonApp, type:

- pip install nifpga

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# pip install nifpga
Collecting nifpga
  Using cached nifpga-22.0.0-py3-none-any.whl.metadata (2.6 kB)
Collecting future (from nifpga)
  Using cached future-1.0.0-py3-none-any.whl.metadata (4.0 kB)
Using cached nifpga-22.0.0-py3-none-any.whl (38 kB)
Using cached future-1.0.0-py3-none-any.whl (491 kB)
Installing collected packages: future, nifpga
Successfully installed future-1.0.0 nifpga-22.0.0
```

# Using FileZilla for File Transfer

## (Manual Way)

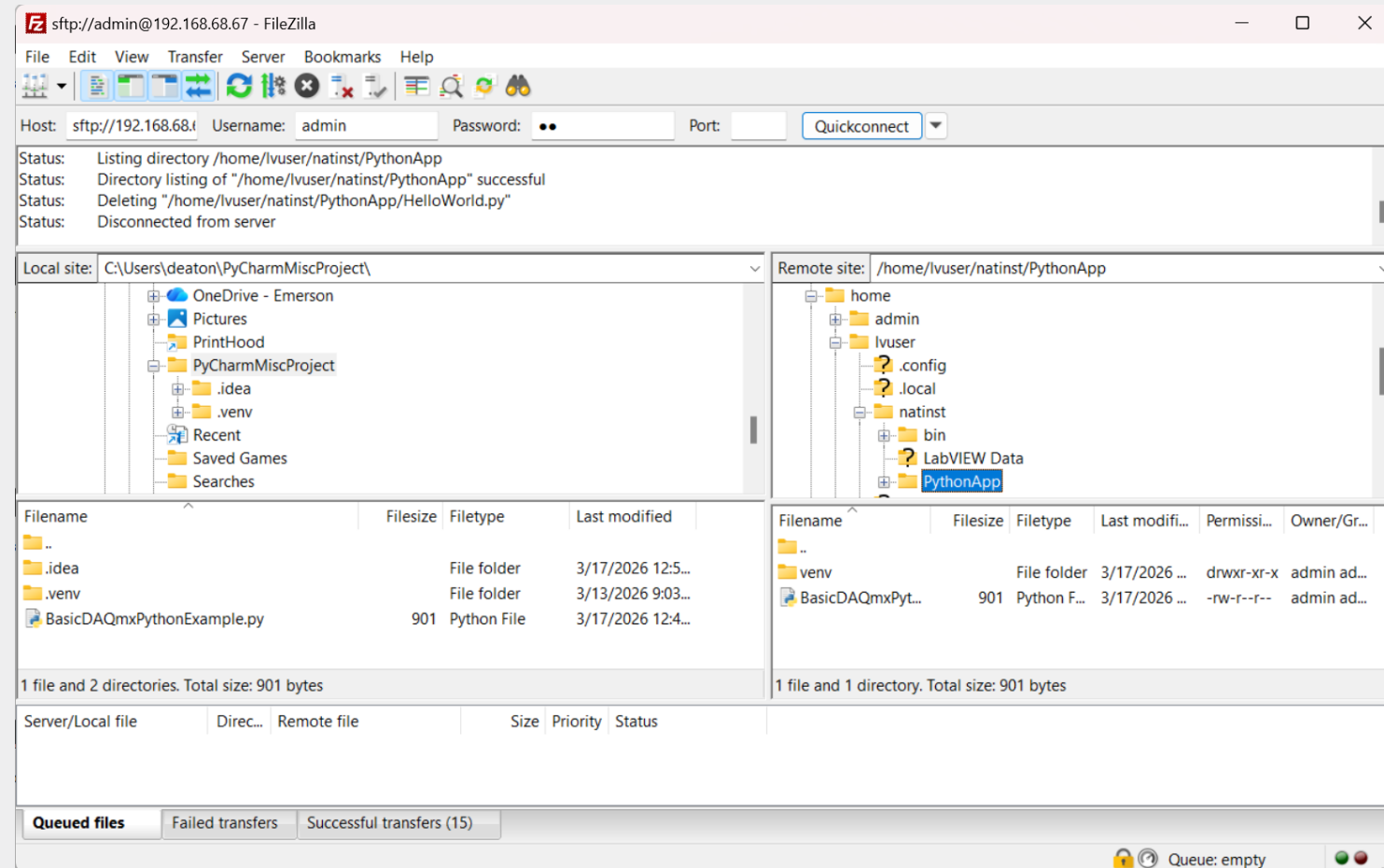
### Set

- Host Name to cRIO IP Address
- Enter Username
- Enter Password
- Port to 22

### Hit Quick Connect

Move python scripts to:

/home/lvuser/PythonApp



# Run Python Code

## (Manual Way)

In ((venv)) /home/lvuser/PythonApp, type:

- `python <scriptname>.py`

Using provided example:

- `python BasicDAQmxPythonExample.py`

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# python BasicDAQmxPythonExample.py
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7fb271058680>
Acquired data: [-1.39289974e-04  2.48684476e-05  2.48684476e-05  2.48684476e-05
 2.48684476e-05  2.48684476e-05 -1.39289974e-04  2.48684476e-05
-1.39289974e-04 -1.39289974e-04  2.48684476e-05  2.48684476e-05
-1.39289974e-04 -1.39289974e-04  2.48684476e-05 -3.03448396e-04
 2.48684476e-05 -1.39289974e-04  2.48684476e-05 -1.39289974e-04
-1.39289974e-04  2.48684476e-05 -1.39289974e-04  2.48684476e-05
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-1.39289974e-04 -1.39289974e-04  2.48684476e-05 -1.39289974e-04
-1.39289974e-04 -3.03448396e-04  2.48684476e-05  2.48684476e-05
-1.39289974e-04  2.48684476e-05  2.48684476e-05  2.48684476e-05
-1.39289974e-04 -1.39289974e-04  2.48684476e-05 -1.39289974e-04
 2.48684476e-05 -1.39289974e-04 -1.39289974e-04  2.48684476e-05
2.48684476e-05 1.89026870e-04 -1.39289974e-04  2.48684476e-05
2.48684476e-05 2.48684476e-05 2.48684476e-05 2.48684476e-05
2.48684476e-05 -1.39289974e-04  2.48684476e-05 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 -3.03448396e-04
2.48684476e-05 -1.39289974e-04  2.48684476e-05  2.48684476e-05
2.48684476e-05 2.48684476e-05 2.48684476e-05 -3.03448396e-04
2.48684476e-05 -3.03448396e-04  2.48684476e-05 -1.39289974e-04]
```

```
1  #!/bin/bash
2
3  echo "Installing Python..."
4  opkg install python3-pip
5  echo "Moving to User Directory..."
6  cd '/home/lvuser/'
7  echo "Creating New Application Directory..."
8  mkdir 'PythonApp'
9  echo "Moving to New Application Directory..."
10 cd 'PythonApp'
11 echo "Create Virtual Environment..."
12 python -m venv venv
13 echo "Activate Virtual Environment..."
14 source venv/bin/activate
15 echo "Upgrading PiP..."
16 python -m pip install --upgrade pip
17 echo "Install NI DAQmx..."
18 pip install nidaqmx
19 echo "Install NI FPGA..."
20 pip install nifpga
```

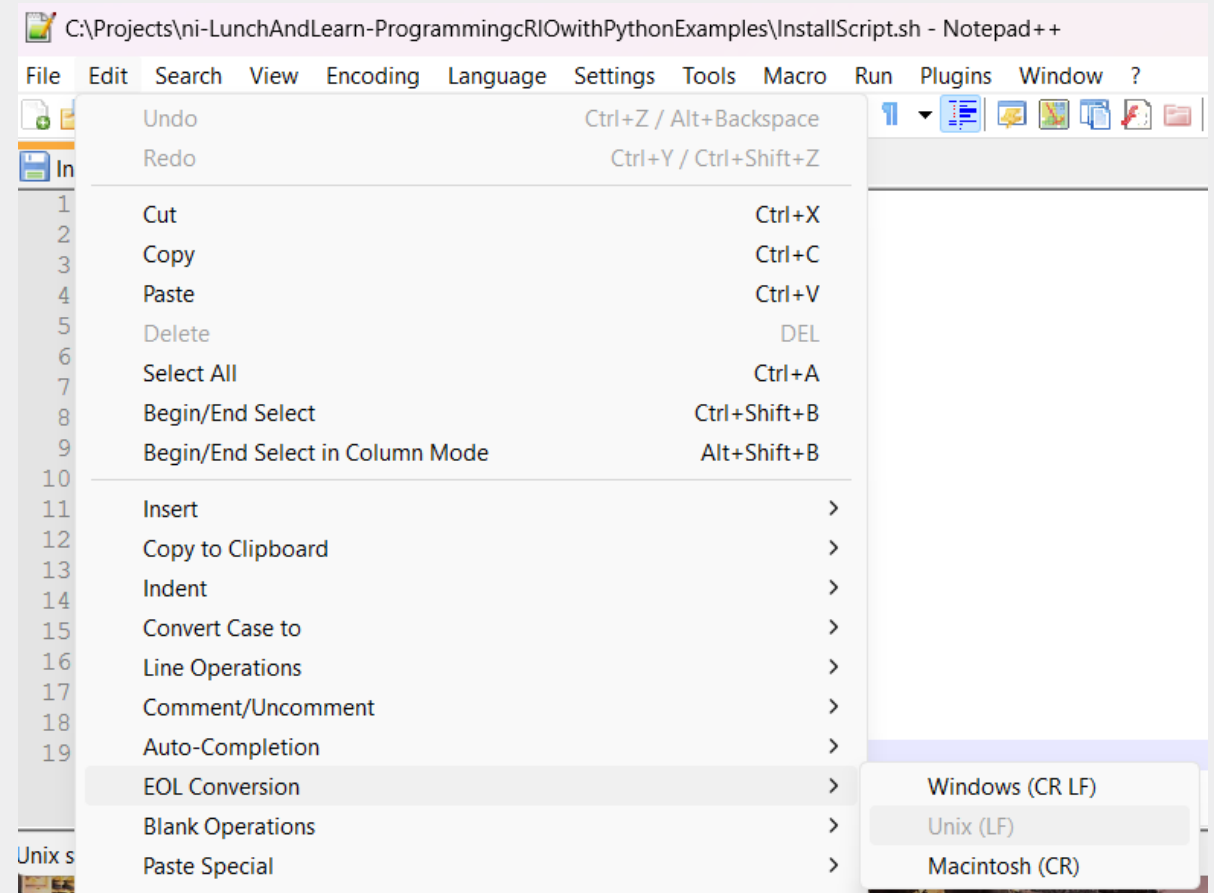
# Demo: Run BASH Script

---

# TIP: Writing a BASH File for NI Linux RT

Extension of .sh

Change EOL Conversion to Unix (LF)



# DEMO: Run BASH Script

Review the BASH file (InstallScript.sh)

- Highlight setting up folder structure
- Highlight creating and activating virtual environment
- Highlight installing nidaqmx and nifpga

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

Move InstallScript.sh to the root/default cRIO directory

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

Login to cRIO via SSH (through PuTTY)

Run the script: (Type: bash InstallScript.sh)

See: Videos\Run BASH Script.mp4

```
1  #!/bin/bash
2
3  echo "Installing Python..."
4  opkg install python3-pip
5  echo "Moving to User Directory..."
6  cd '/home/lvuser/'
7  echo "Creating New Application Directory..."
8  mkdir 'PythonApp'
9  echo "Moving to New Application Directory..."
10 cd 'PythonApp'
11 echo "Create Virtual Environment..."
12 python -m venv venv
13 echo "Activate Virtual Environment..."
14 source venv/bin/activate
15 echo "Upgrading PiP..."
16 python -m pip install --upgrade pip
17 echo "Install NI DAQmx..."
18 pip install nidaqmx
19 echo "Install NI FPGA..."
20 pip install nifpga
```

# DEMO: Run BASH Script Summary

```
1  #!/bin/bash
2
3  echo "Installing Python..."
4  opkg install python3-pip
5  echo "Moving to User Directory..."
6  cd '/home/lvuser/natinst/'
7  echo "Creating New Application Directory..."
8  mkdir 'PythonApp'
9  echo "Moving to New Application Directory..."
10 cd 'PythonApp'
11 echo "Create Virtual Environment..."
12 python -m venv venv
13 echo "Activate Virtual Environment..."
14 source venv/bin/activate
15 echo "Upgrading PiP..."
16 python -m pip install --upgrade pip
17 echo "Install NI DAQmx..."
18 pip install nidagmx
19 pip install nifpga
```



Run BASH files

Create a dedicated folder for Python code

Create a virtual environment & activate

Install NI DAQmx and NI FPGA Python packages

Ultimately, cRIO in (NI) Linux (RT)



```
BasicDAQmxPythonExample.py x
33
34 with nidaqmx.Task() as task:
35     # Add an analog input voltage channel
36     task.ai_channels.add_ai_voltage_chan(f"{mod_name}/{chan_name}")
37
38     # Configure simple timing task
39     task.timing.cfg_samp_clk_timing(rate=sample_rate, samps_per_chan=num_samples, sample_mode=AcquisitionType.FINITE)
40
41     # Allocate buffer
42     data = np.zeros(num_samples, dtype=np.float64)
43     reader = AnalogSingleChannelReader(task.in_stream)
44
45     # Start and read data
46     task.start()
47     reader.read_many_sample(data, number_of_samples_per_channel=num_samples, timeout=10.0)
48
```

# Demo: Python + DAQmx on cRIO

---

# TIP: Build skills...

For the main script, tell AI to add:

```
import locale
locale.setlocale(locale.LC_ALL, 'en_US.UTF-8')
```

← Skills

×

compactrio-python ⓘ

by You

Overview

Contents · 1

Description ⓘ

Use this skill whenever the user asks to create, write, or generate a Python program or script targeting an NI CompactRIO (cRIO) controller. NI CompactRIO Python programs must always import the locale module and explicitly set the locale to 'en\_US.UTF-8' right after the imports, otherwise number/string formatting on the controller can misbehave. Trigger this skill even if the user only says "cRIO," "CompactRIO," or "NI controller" without spelling out the full requirement — the locale setup is mandatory boilerplate for every such program.

Slash command

/compactrio-python

# DEMO: Python + DAQmx on cRIO

Review NI DAQmx Python Examples\BasicDAQmxPythonExample.py

- Highlight cRIO hardware (modules being used)
- Highlight import nidaqmx
- Highlight nidaqmx functions

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

- Move BasicDAQmxPythonExample.py to /home/lvuser/PythonApp

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

- Change Directory: `cd '/home/lvuser/PythonApp'`
- Activate Virtual Env: `source venv/bin/activate`
- Run Python Script: `python BasicDAQmxPythonExample.py`

See: Videos\Python + DAQmx on cRIO.mp4

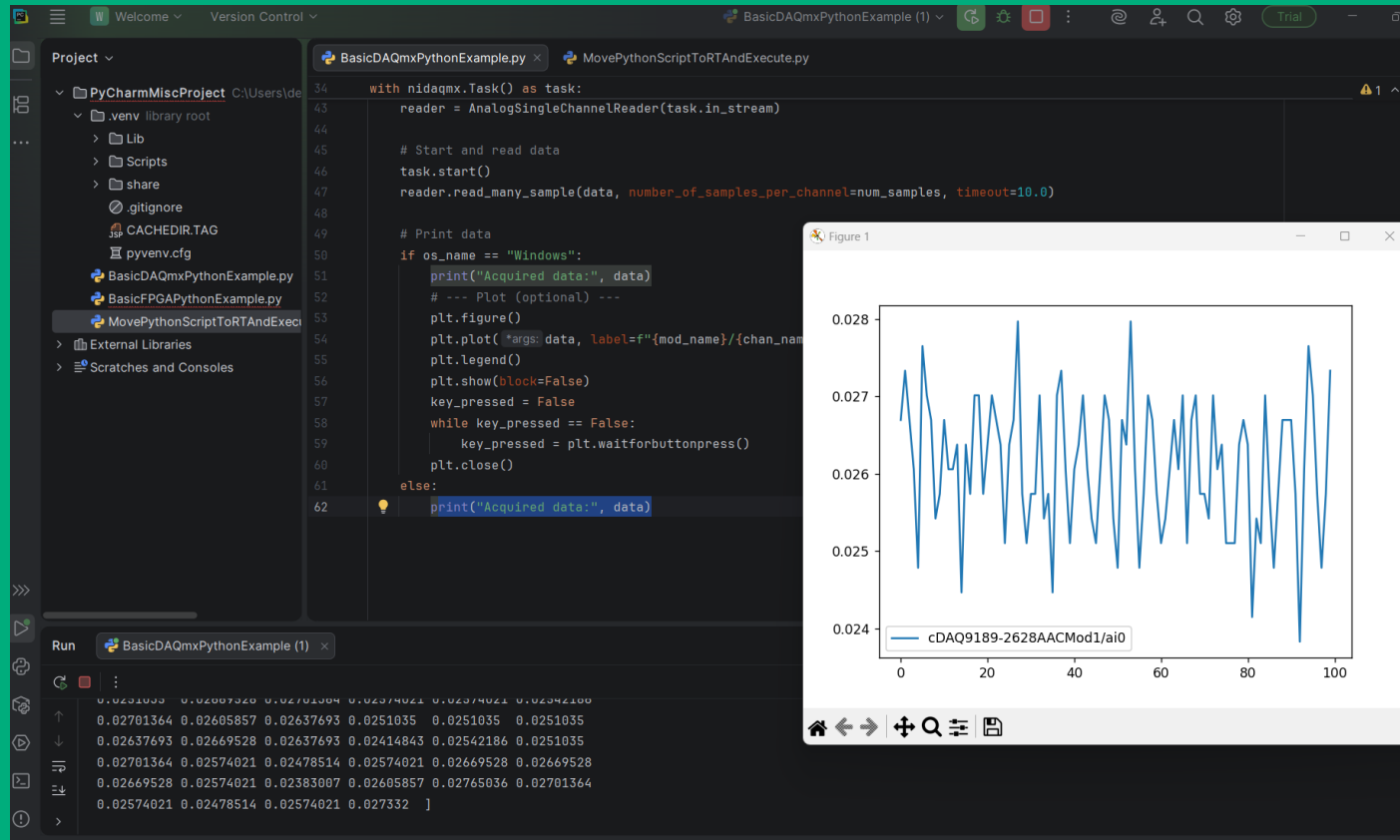
```
192.168.68.67 - PuTTY
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
End of banner message from server
Last login: Wed Mar 18 14:21:58 2026 from 192.168.68.64
admin@NI-cRIO-9045-01CEEDF1:~# cd '/home/lvuser/natinst/PythonApp'
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp# python BasicDAQmxPythonExample.py
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f2592ec0ae0>
Acquired data: [ 1.89026870e-04 -1.39289974e-04 -3.03448396e-04 -3.03448396e-04
2.48684476e-05 2.48684476e-05 2.48684476e-05 2.48684476e-05
2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -1.39289974e-04 -3.03448396e-04
-1.39289974e-04 1.89026870e-04 -1.39289974e-04 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 2.48684476e-05 2.48684476e-05
-3.03448396e-04 -1.39289974e-04 -1.39289974e-04 2.48684476e-05
-1.39289974e-04 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 2.48684476e-05 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 -1.39289974e-04 1.89026870e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -3.03448396e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 2.48684476e-05
-1.39289974e-04 2.48684476e-05 1.89026870e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 2.48684476e-05
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 -3.03448396e-04
-3.03448396e-04 -1.39289974e-04 2.48684476e-05 -3.03448396e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 1.89026870e-04 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 -3.03448396e-04
-3.03448396e-04 -1.39289974e-04 2.48684476e-05 -3.03448396e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 1.89026870e-04
2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -1.39289974e-04]
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp#
```

# DEMO: Python + DAQmx on cRIO Summary

Run Python on cRIO

Use NI DAQmx API

```
192.168.68.67 - PuTTY
login as: admin
Pre-authentication banner message from server:
| NI Linux Real-Time (run mode)
|
| End of banner message from server
Last login: Wed Mar 18 14:21:58 2026 from 192.168.68.64
admin@NI-cRIO-9045-01CEEDF1:~$ cd '/home/lvuser/natinst/PythonApp'
admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp$ source venv/bin/activate
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp$ python BasicDAQmxPythonExample.py
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f2592ec0ae0>
Acquired data: [ 1.89026870e-04 -1.39289974e-04 -3.03448396e-04 -3.03448396e-04
 2.48684476e-05 2.48684476e-05 2.48684476e-05 2.48684476e-05
 2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -3.03448396e-04
-1.39289974e-04 1.89026870e-04 -1.39289974e-04 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 2.48684476e-05 2.48684476e-05
-1.39289974e-04 -1.39289974e-04 -1.39289974e-04 2.48684476e-05
-1.39289974e-04 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 2.48684476e-05 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 -1.39289974e-04 1.89026870e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -3.03448396e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 2.48684476e-05
-1.39289974e-04 2.48684476e-05 1.89026870e-04 -1.39289974e-04
2.48684476e-05 2.48684476e-05 -1.39289974e-04 2.48684476e-05
-3.03448396e-04 -1.39289974e-04 2.48684476e-05 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 2.48684476e-05 -1.39289974e-04
2.48684476e-05 2.48684476e-05 1.89026870e-04 -1.39289974e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 2.48684476e-05
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 -3.03448396e-04
-3.03448396e-04 -1.39289974e-04 2.48684476e-05 -3.03448396e-04
2.48684476e-05 -1.39289974e-04 2.48684476e-05 -1.39289974e-04
-1.39289974e-04 2.48684476e-05 -1.39289974e-04 1.89026870e-04
2.48684476e-05 -1.39289974e-04 -1.39289974e-04 -1.39289974e-04]
((venv) ) admin@NI-cRIO-9045-01CEEDF1:/home/lvuser/natinst/PythonApp$
```



# Demo: Run Python on Windows w/ cDAQ (DAQmx Only)

# TIP...Run with cDAQ on Windows First...(DAQmx Only)

## Easiest Debug

```
os_name = platform.system()
if os_name == "Windows":
    import matplotlib.pyplot as plt

#Highlight NI DAQmx works on Windows and on NI Linux RT
print(f"{Path(__file__).name} is executing on {os_name}")
if os_name == "Windows":
    #Set Module Names for the debug system
    mod_name = "cDAQ9189-2628AACMod1"
elif os_name == "Linux":
    #Set Module Names for the cRIO
    mod_name = "Mod4"
```

```
# Print data
if os_name == "Windows":
    print("Acquired data:", data)
    # --- Plot (optional) ---
    plt.figure()
    plt.plot(*args: data, label=f"{mod_name}/{chan_name}")
    plt.legend()
    plt.show(block=False)
    key_pressed = False
    while key_pressed == False:
        key_pressed = plt.waitforbuttonpress()
    plt.close()
else:
    print("Acquired data:", data)
```

# Debugging (DAQmx only)

cRIO is an embedded target (Limited GUI)

Windows is easier for debugging

Build cases for running the same code on Windows as on NI Linux RT

- Import based on OS
- GUI Plotting based on OS

Debug most of the code on Windows

Use a cDAQ 9189 chassis instead of cRIO chassis

- Use same NI DAQmx API
- Use C Series Modules
- Use same wiring

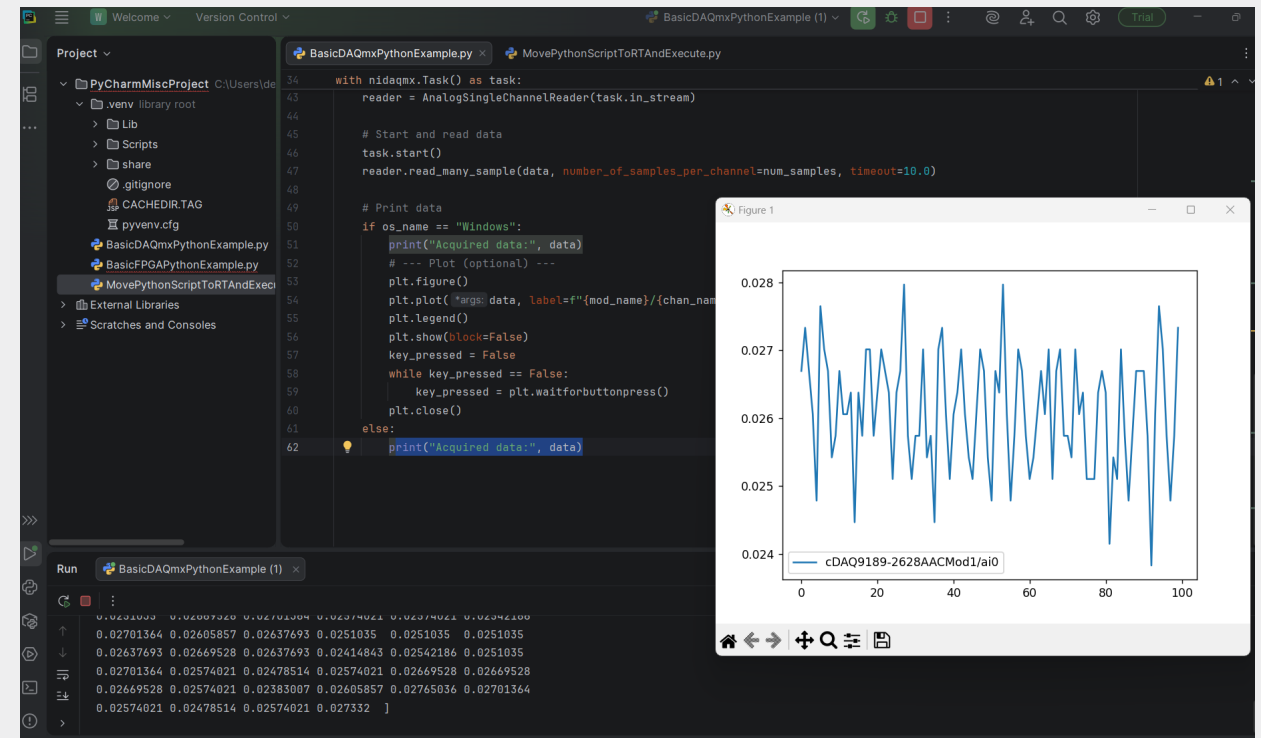
# Debugging (DAQmx only)

Review NI DAQmx Python Examples\BasicDAQmxPythonExample.py

- Highlight OS based cases
  - Import
  - Plotting

Run BasicDAQmxPythonExample.py on Windows

See: Videos\Run Python on Windows with cDAQ\_DAQmx Only.mp4

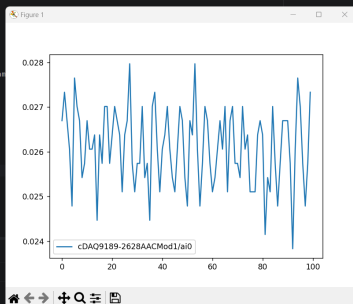






## Only swap cRIO Chassis for cDAQ Chassis

## Configuration changes slightly, main code UNTOUCHED



```
Run MovePythonScriptToRTAndExecute x
" C:\Users\deaton\PycharmProjects\cRIO Python DAQmx Example Program\.venv\Scripts\python.exe" "C:\Users\deaton\PycharmProjects\cRIO Python DAQmx
Running on: Windows
Remote command: cd /home/lvuser/PythonApp/ && /home/lvuser/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.57 using password...
Transferring 'C:\Users\deaton\PycharmProjects\cRIO Python DAQmx Example Program\BasicDAQmxPythonExample.py' -> '192.168.68.57:/home/lvuser/Pyth
Transfer verified. File size: 1885 bytes
Connected to 192.168.68.57

Changing to directory: /home/lvuser/PythonApp/
Executing script: BasicDAQmxPythonExample.py

-----
Output:
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f402d8785e0>
Acquired data: [ 0.4207632  0.41665923  0.4207632  0.41731587  0.4207632  0.4179725
 0.42027073  0.41912161  0.41846498 -0.42350359 -0.42432438 -0.42301111
-0.42481685 -0.42219032 -0.42596596 -0.42120537 -0.4258018  -0.421862
-0.42563764 -0.42120537 -0.4258018  -0.42169785 -0.42613012 -0.42169785
-0.42596596 -0.42268279 -0.42448854 -0.42350359 -0.42317527 -0.42416022
-0.42317527 -0.42481685 -0.42301111 -0.42563764 -0.42251864 -0.4258018
-0.42219032 -0.42596596 -0.421862  -0.42629428 -0.42120537 -0.42727922
-0.42104121 -0.4258018  -0.42169785 -0.4258018  -0.42284695 -0.42432438
-0.42498101 -0.42301111 -0.42514517 -0.42268279 -0.42498101 -0.42251864
-0.42498101 -0.42268279 -0.42613012 -0.421862  -0.42678675 -0.4207129
```

# Demo: Automate Remote Workflow

# Remote Development Options

## Windows → NI Linux RT

Some IDEs allow automatic file sync and SSH connection

Use a python script to transfer and run your application

# Demo: Automate Remote Workflow

Review NI DAQmx Python Examples\MovePythonScriptToRTAndExecute.py

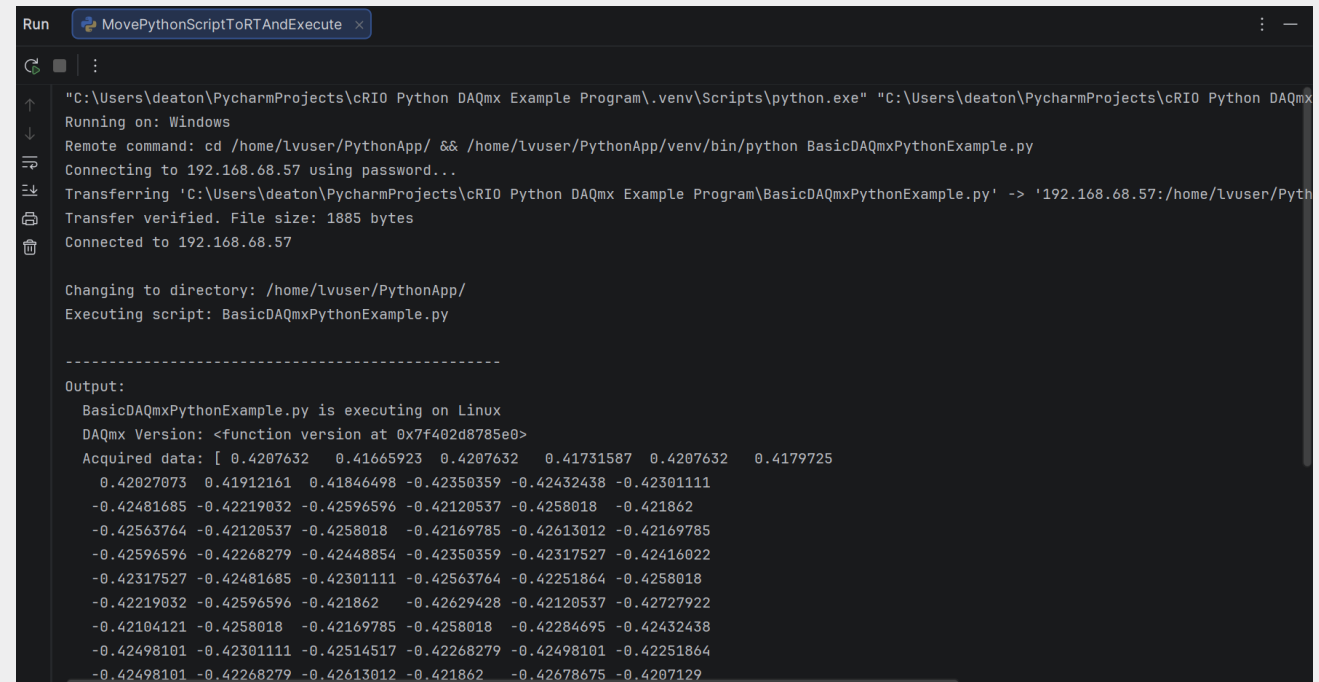
- Highlight paramiko package
- Highlight SSH
- Highlight sFTP
- Highlight remote command
- Highlight streaming back response

In FileZilla

- Connect to cRIO
- GoTo: /home/lvuser/PythonApp/
- Delete file BasicDAQmxPythonExample.py

Run MovePythonScriptToRTAndExecute.py

- Show Result
- Show File back on cRIO



```
Run MovePythonScriptToRTAndExecute x
C:\Users\deaton\PycharmProjects\cRIO Python DAQmx Example Program\.venv\Scripts\python.exe "C:\Users\deaton\PycharmProjects\cRIO Python DAQmx Example Program\BasicDAQmxPythonExample.py"
Running on: Windows
Remote command: cd /home/lvuser/PythonApp/ && /home/lvuser/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.57 using password...
Transferring 'C:\Users\deaton\PycharmProjects\cRIO Python DAQmx Example Program\BasicDAQmxPythonExample.py' -> '192.168.68.57:/home/lvuser/PythonApp/'
Transfer verified. File size: 1885 bytes
Connected to 192.168.68.57

Changing to directory: /home/lvuser/PythonApp/
Executing script: BasicDAQmxPythonExample.py

-----
Output:
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f402d8785e0>
Acquired data: [ 0.4207632  0.41665923  0.4207632  0.41731587  0.4207632  0.4179725
 0.42027073  0.41912161  0.41846498 -0.42350359 -0.42432438 -0.42301111
-0.42481685 -0.42219032 -0.42596596 -0.42120537 -0.4258018  -0.421862
-0.42563764 -0.42120537 -0.4258018  -0.42169785 -0.42613012 -0.42169785
-0.42596596 -0.42268279 -0.42448854 -0.42350359 -0.42317527 -0.42416022
-0.42317527 -0.42481685 -0.42301111 -0.42563764 -0.42251864 -0.4258018
-0.42219032 -0.42596596 -0.421862  -0.42629428 -0.42120537 -0.42727922
-0.42104121 -0.4258018  -0.42169785 -0.4258018  -0.42284695 -0.42432438
-0.42498101 -0.42301111 -0.42514517 -0.42268279 -0.42498101 -0.42251864
-0.42498101 -0.42268279 -0.42613012 -0.421862  -0.42678675 -0.4207129]
```

See: Videos\Automate Remote Workflow.mp4

# DEMO: Automate Remote Workflow Summary

```
Run MovePythonScriptToRTAndExecute x
C:\Users\deaton\PycharmProjects\cRIO_Python_DAQmx_Example_Program\.venv\Scripts\python.exe "C:\Users\deaton\PycharmProjects\cRIO_Python_DAQmx_Example_Program\BasicDAQmxPythonExample.py"
Running on: Windows
Remote command: cd /home/lvuser/PythonApp/ && /home/lvuser/PythonApp/venv/bin/python BasicDAQmxPythonExample.py
Connecting to 192.168.68.57 using password...
Transferring 'C:\Users\deaton\PycharmProjects\cRIO_Python_DAQmx_Example_Program\BasicDAQmxPythonExample.py' -> '192.168.68.57:/home/lvuser/PythonApp/'
Transfer verified, File size: 1865 bytes
Connected to 192.168.68.57

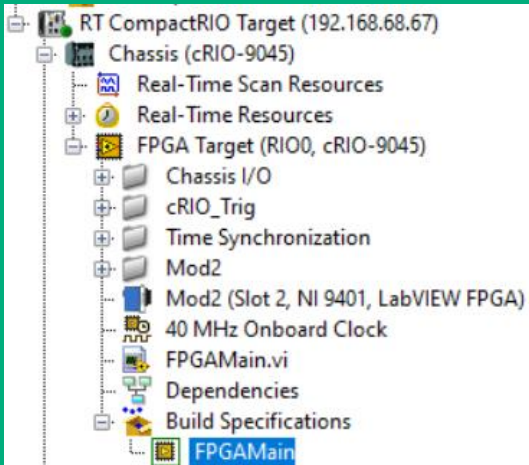
Changing to directory: /home/lvuser/PythonApp/
Executing script: BasicDAQmxPythonExample.py

-----
Output:
BasicDAQmxPythonExample.py is executing on Linux
DAQmx Version: <function version at 0x7f402d8785e0>
Acquired data: [ 0.4207632  0.41665923  0.4207632  0.41731587  0.4207632  0.4179725
 0.42077073  0.41917161  0.41846498 -0.42350359 -0.42432438 -0.42301111
-0.42481085 -0.42219032 -0.42596596 -0.42128037 -0.42580818 -0.42169785
-0.42543764 -0.42128037 -0.42580818 -0.42169785 -0.42613012 -0.42169785
-0.42596596 -0.42268279 -0.42448854 -0.42350359 -0.42317527 -0.42416022
-0.42317527 -0.42481085 -0.42301111 -0.42543764 -0.42251864 -0.42580818
-0.42219032 -0.42596596 -0.421862 -0.42629428 -0.42128037 -0.42727922
-0.42184121 -0.42580818 -0.42169785 -0.42580818 -0.42284695 -0.42432438
-0.42498101 -0.42301111 -0.42514517 -0.42268279 -0.42498101 -0.42251864
-0.42498101 -0.42268279 -0.42613012 -0.421862 -0.42678675 -0.4207129
```



Automatically:

- Move Files to RT
- Run Script on RT
- All Through Windows IDE



```
((venv) ) admin@NI-CRIO-9047:/home/lvuser/PythonApp# python BasicFPGAPythonExample.py  
Connecting to FPGA: RIO0  
Bitfile: /home/lvuser/PythonApp/FPGAMainBitfile_9047.lvbitx  
FPGA is running.
```

| Timestamp | Loop# | Count     | Simulated Mod2 DIO0         |
|-----------|-------|-----------|-----------------------------|
| 19:15:44  | 1     | Count = 1 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 2     | Count = 1 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 3     | Count = 2 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 4     | Count = 2 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 5     | Count = 3 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 6     | Count = 3 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 7     | Count = 4 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 8     | Count = 4 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 9     | Count = 5 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 10    | Count = 5 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 11    | Count = 6 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 12    | Count = 6 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 13    | Count = 7 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 14    | Count = 7 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 15    | Count = 8 | Simulated Mod2 DIO0 = True  |
| 19:15:46  | 16    | Count = 8 | Simulated Mod2 DIO0 = False |

# Demo: Python + FPGA on cRIO

# Demo: Python + FPGA on cRIO

## Prerequisites

Create a LabVIEW Project with your cRIO

Add the FPGA Target

Move Only Desired Modules under the FPGA (PER MODULE)

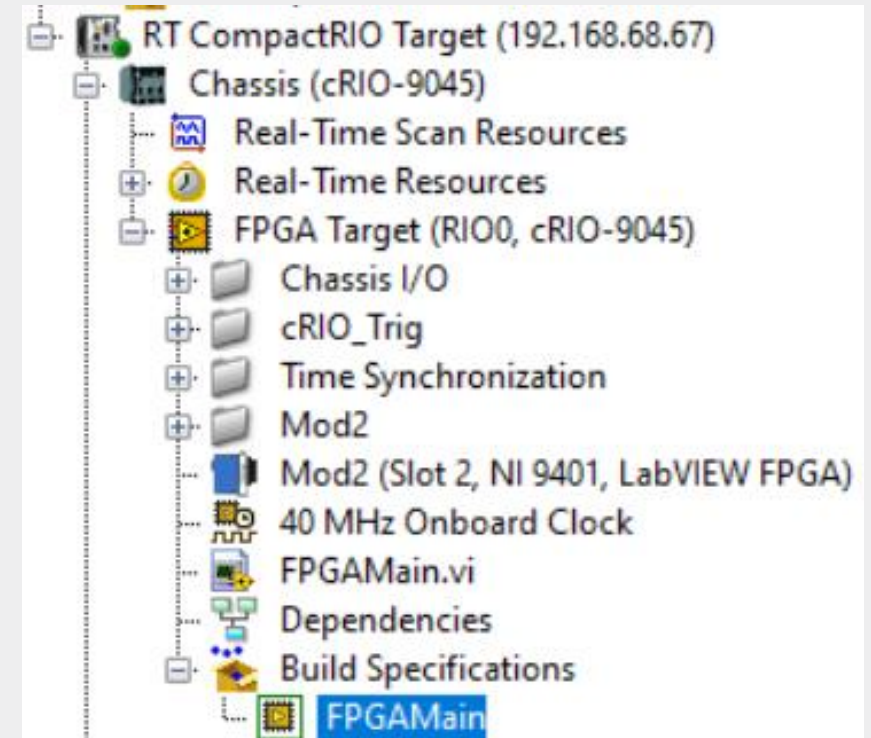
Create your LabVIEW FPGA Program

(Test your Code in Simulated Mode)

Compile your program into a bitfile

(Test your Code w/ Hardware)

Then, write your Python...



# Demo: Automate Remote Workflow

Review NI FPGA Python Examples\BasicFPGAPythonExample.py

- Highlight import nifpga
- Highlight creating nifpga session
- Highlight starting/reseting fpga
- Highlight reading indicators /writing controls

Use FileZilla to connect to the cRIO (Host = cRIO IP, port 22)

- Move BasicFPGAPythonExample.py to /home/lvuser/PythonApp

Use PuTTY to connect to the cRIO (Host = cRIO IP, port 22)

- Change Directory: `cd '/home/lvuser/PythonApp'`
- Activate Virtual Env: `source venv/bin/activate`
- Run Python Script: `python BasicFPGAPythonExample.py`

See: Videos\Python + FPGA on cRIO.mp4

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# python BasicFPGAPythonExample.py
Connecting to FPGA: RIO0
Bitfile: /home/lvuser/PythonApp/FPGAMainBitfile_9047.lvbitx
FPGA is running.
```

| Timestamp | Loop# | Count     | Simulated Mod2 DIO0         |
|-----------|-------|-----------|-----------------------------|
| 19:15:44  | 1     | Count = 1 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 2     | Count = 1 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 3     | Count = 2 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 4     | Count = 2 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 5     | Count = 3 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 6     | Count = 3 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 7     | Count = 4 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 8     | Count = 4 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 9     | Count = 5 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 10    | Count = 5 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 11    | Count = 6 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 12    | Count = 6 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 13    | Count = 7 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 14    | Count = 7 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 15    | Count = 8 | Simulated Mod2 DIO0 = True  |
| 19:15:46  | 16    | Count = 8 | Simulated Mod2 DIO0 = False |



# DEMO: Automate Remote Workflow Summary



Use the cRIO FPGA in Python (PER MODULE)

```
((venv) ) admin@NI-cRIO-9047:/home/lvuser/PythonApp# python BasicFPGAPythonExample.py
Connecting to FPGA: RIO0
Bitfile: /home/lvuser/PythonApp/FPGAMainBitfile_9047.lvbitx
FPGA is running.
```

| Timestamp | Loop# | Count     | Simulated Mod2 DIO0         |
|-----------|-------|-----------|-----------------------------|
| 19:15:44  | 1     | Count = 1 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 2     | Count = 1 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 3     | Count = 2 | Simulated Mod2 DIO0 = True  |
| 19:15:44  | 4     | Count = 2 | Simulated Mod2 DIO0 = False |
| 19:15:44  | 5     | Count = 3 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 6     | Count = 3 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 7     | Count = 4 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 8     | Count = 4 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 9     | Count = 5 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 10    | Count = 5 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 11    | Count = 6 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 12    | Count = 6 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 13    | Count = 7 | Simulated Mod2 DIO0 = True  |
| 19:15:45  | 14    | Count = 7 | Simulated Mod2 DIO0 = False |
| 19:15:45  | 15    | Count = 8 | Simulated Mod2 DIO0 = True  |
| 19:15:46  | 16    | Count = 8 | Simulated Mod2 DIO0 = False |

# Demo: Full Application

---

# Demo: Full Application

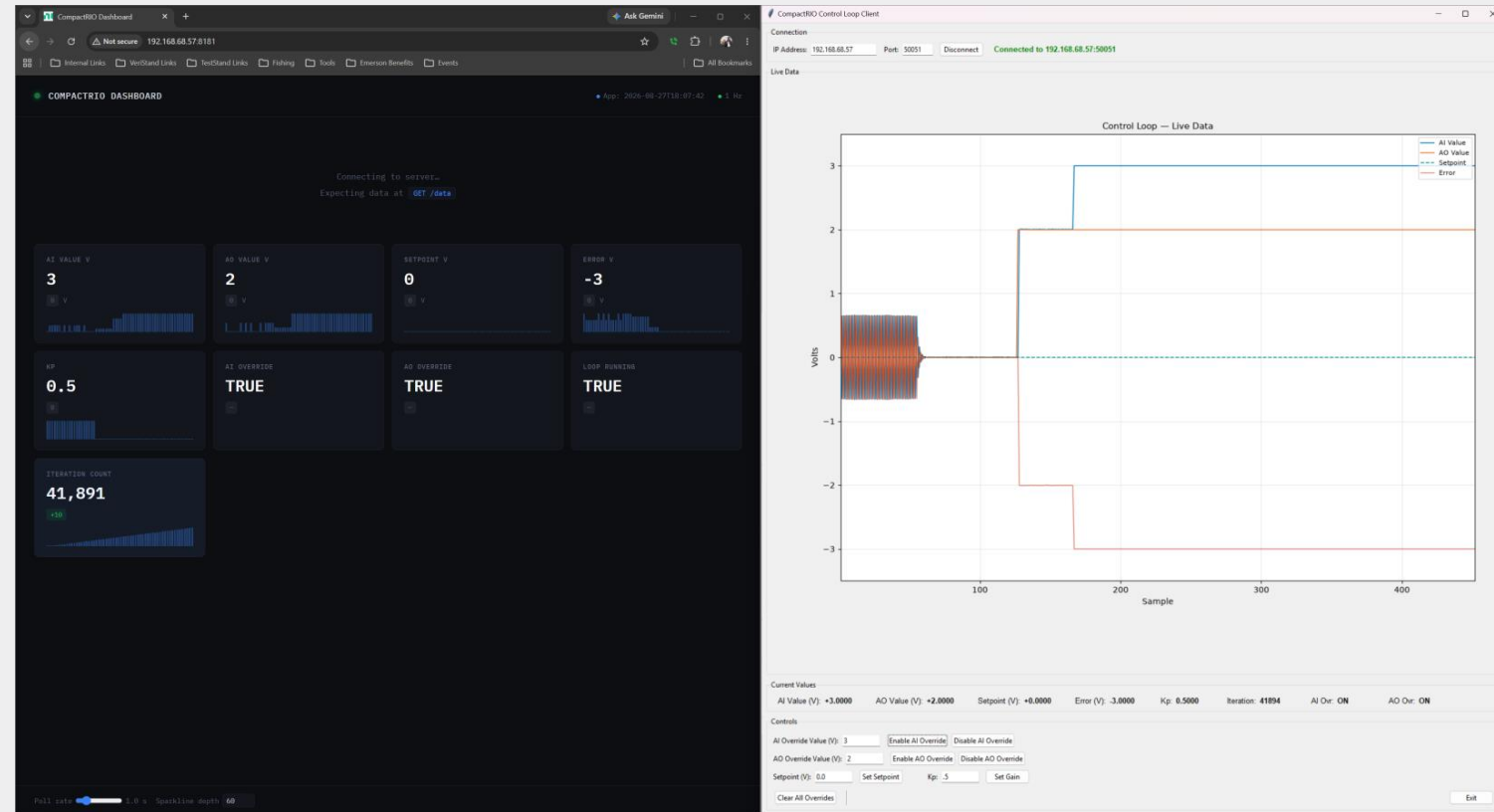
Vibe coded:

- cRIO Control App
- Locally hosted webpage
- gRPC Interface
- Windows gRPC GUI

Already:

- Installed drivers / python / venv
- Moved application to cRIO
- Validated app function

Now need set as executable / run on reboot...



# Demo: Full Application

## Review Code

control\_loop.proto → Communication Definition

\*\_pb2.py & \*\_pb2\_grpc.py → Communication API generated from Proto

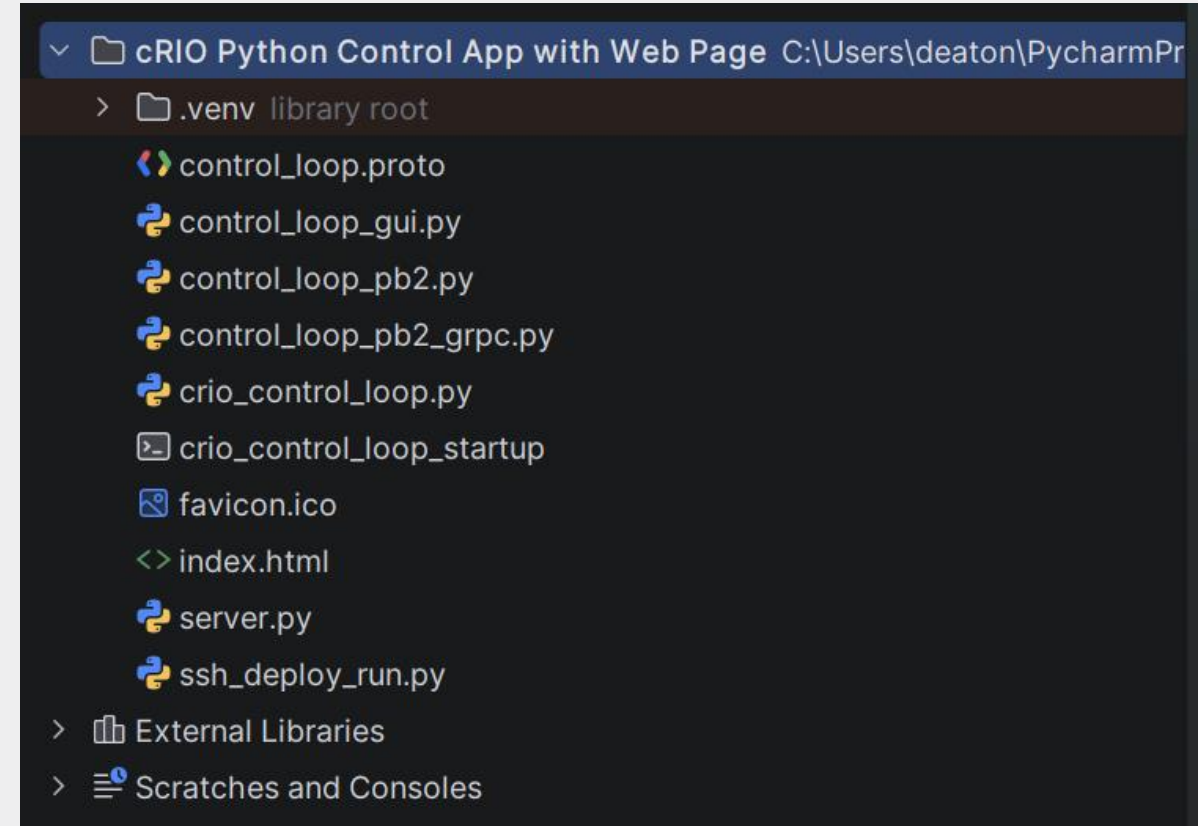
crio\_control\_loop.py → Main App

- Calls server.py
- Calls index.html
- Calls favicon.ico

crio\_control\_loop\_startup → startup script for SysV

ssh\_deploy\_run.py → Host app moves crio files and remotely executes

control\_loop\_gui.py → Host gRPC GUI



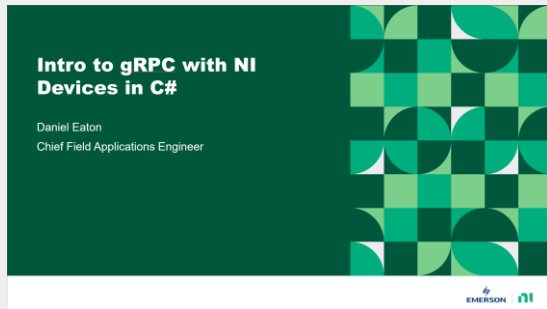
# gRPC Lunch and Learn Available!!

Want more gRPC?!?

LabVIEW gRPC: <https://github.com/NISystemsEngineering/ni-LunchAndLearn-LabVIEWgRPCFundamentalsExamples>



NI Drivers gRPC: <https://github.com/NISystemsEngineering/ni-LunchAndLearn-gRPCDeviceOverviewExamples>



# Demo: Full Application

Pro Tip: Add simple wait to start of app...

cRIO drivers need time to initialize...

Write code that checks if drivers are ready or...wait

```
def main() -> None: 1 usage
    time.sleep(30)
    logging.basicConfig(
        level=logging.INFO,
        format="%(asctime)s %(levelname)-8s %(message)s",
    )
```

# Demo: Full Application

## Running from Windows

(assumes venv is setup)

From Windows execute:

Run command (in Setup Info.txt)

- Files are moved to cRIO
- Application started remotely
- Wait (30 seconds)

Connect to the webpage in Chrome

- IPAddress:8181

Start the Python gRPC Client:

- control\_loop\_gui.py

In terminal: Ctrl C stops

```
(.venv) PS C:\Users\deaton\PycharmProjects\cRIO Python Control App with Web Page> python ssh_deploy_run.py --host 192.168.68.57 --user admin --files control_loop_
crio_control_loop.py
Password for admin@192.168.68.57:
Connected to admin@192.168.68.57:22

Transferring files _
control_loop_pb2.py -> /home/lvuser/cricontrolapp/control_loop_pb2.py (3,595 bytes) _ OK
control_loop_pb2_grpc.py -> /home/lvuser/cricontrolapp/control_loop_pb2_grpc.py (14,571 bytes) _ OK
crio_control_loop.py -> /home/lvuser/cricontrolapp/crio_control_loop.py (19,335 bytes) _ OK
server.py -> /home/lvuser/cricontrolapp/server.py (4,317 bytes) _ OK
index.html -> /home/lvuser/cricontrolapp/index.html (14,661 bytes) _ OK
favicon.ico -> /home/lvuser/cricontrolapp/favicon.ico (4,286 bytes) _ OK

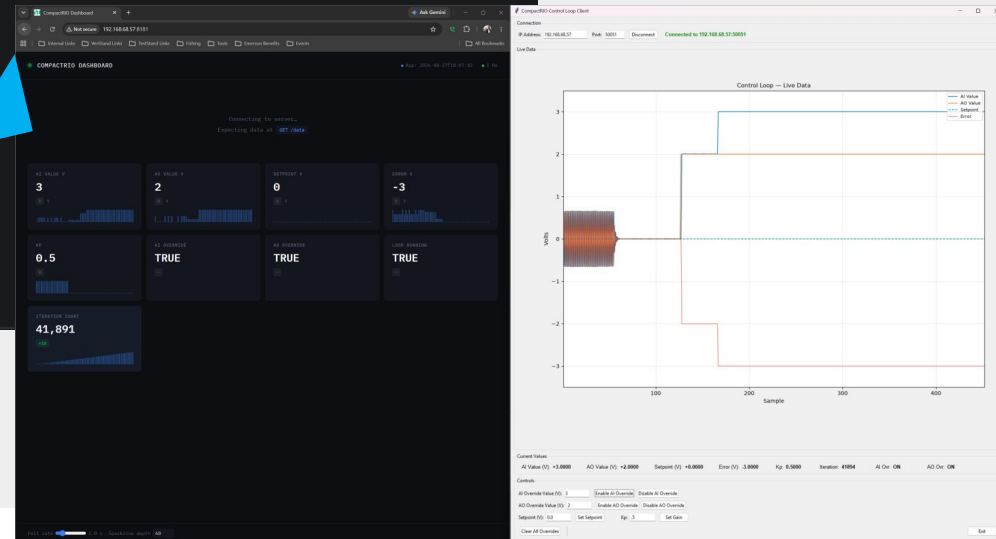
All files uploaded to /home/lvuser/cricontrolapp

Executing: cd /home/lvuser/cricontrolapp && source /home/lvuser/cricontrolapp/venv/bin/activate && python3 -u /home/lvuser/cricontrolapp/crio_control_loop.py
Press Ctrl-C to stop.

2026-08-27 19:23:56,476 INFO Launching dashboard server: /home/lvuser/cricontrolapp/venv/bin/python3 /home/lvuser/cricontrolapp/server.py --port 8181
2026-08-27 19:23:56,494 INFO gRPC server listening on [::]:50051
Dashboard running at http://192.168.68.57:8181
Reading variables from /tmp/crio_variables.json
2026-08-27 19:23:56,755 INFO AI task created on Mod5/ai0 (differential, on-demand)
2026-08-27 19:23:56,764 INFO AO task created on Mod3/ao0 (on-demand)
2026-08-27 19:23:57,089 INFO Control loop running at 10.0 Hz

>>> Ctrl-C caught - sending interrupt to remote process ...
^C
Shutting down.
2026-08-27 19:24:03,972 INFO Caught signal 2 - shutting down...
2026-08-27 19:24:03,996 INFO Control loop stopped; tasks closed.
2026-08-27 19:24:04,002 INFO Stopping dashboard server (pid 18517)...
2026-08-27 19:24:04,006 INFO Dashboard server stopped.
2026-08-27 19:24:04,006 INFO Server stopped.

Remote process exited with code 0
SSH connection closed.
```



# Demo: Full Application

## Create Startup Script

cRIO uses SysV

Create a bash file (no file extension needed)

- Shebang (#!/bin/bash) at the top
- Set python venv path
- Use Start-Stop-Daemon to launch
- Use PID (and pid file) to stop
- Use Log file to record standard outputs
- Use Linux EOL (if on Windows)

Example:

..\cRIO Control App with Web Page\Python Source\crio\_control\_loop\_startup

```
1  #!/bin/bash
2
3  NAME="crio_control_loop_startup"
4  APP_DIR="/home/lvuser/criocontrolapp"
5  PYTHON_VENV="/home/lvuser/criocontrolapp/venv/bin/python"
6  SCRIPT="/home/lvuser/criocontrolapp/crio_control_loop.py"
7  LOG="/var/log/crio_control_loop_startup.log"
8  PIDFILE="/var/run/crio_control_loop_startup.pid"
9
10 do_start() {
11     echo "Starting $NAME"
12     export PYTHONUNBUFFERED=1
13     sleep 2
14     cd "$APP_DIR" || exit 1
15     start-stop-daemon --start --background --no-close --pidfile "$PIDFILE" --make-pidfile --chdir "$APP_DIR" --exec "$PYTHON_VENV" -- "$SCRIPT" $DAEMON_ARGS >> "$LOG" 2>&1
16 }
17
18 do_stop() {
19     echo "Stopping $NAME"
20     start-stop-daemon --stop --pidfile "$PIDFILE" --retry 10 --verbose
21     rm -f "$PIDFILE"
22 }
23
24 do_restart() {
25     do_stop
26     sleep 1
27     do_start
28 }
29
30 case "$1" in
31     start)
32         do_start
33         ;;
34     stop)
35         do_stop
36         ;;
37     restart)
38         do_restart
39         ;;
40     *)
41         echo "Usage: $0 {start|stop|restart}"
42         exit 1
43         ;;
44 esac
45
46 exit 0
```



# Demo: Full Application

## Register Startup Script

Move file to cRIO: /etc/init.d

Set cRIO file permissions:

- `chmod +x /etc/init.d/crio_control_loop_startup`

Test bash file

- `/etc/init.d/crio_control_loop_startup start`
- `/etc/init.d/crio_control_loop_startup stop`

Make bash file part of start up sequence:

- `/usr/sbin/update-rc.d -f crio_control_loop_startup defaults`

Reboot

NOTE: Stopping startup script: `/usr/sbin/update-rc.d -f crio_control_loop_startup remove`

```
admin@NI-cRIO-9047:~# /etc/init.d/crio_control_loop_startup start
Starting crio_control_loop_startup
admin@NI-cRIO-9047:~# /etc/init.d/crio_control_loop_startup stop
Stopping crio_control_loop_startup
Stopped process in pidfile '/var/run/crio_control_loop_startup.pid' (pid 26305).
admin@NI-cRIO-9047:~# /usr/sbin/update-rc.d -f crio_control_loop_startup defaults
Adding system startup for /etc/init.d/crio_control_loop_startup.
admin@NI-cRIO-9047:~# /usr/sbin/update-rc.d -f crio_control_loop_startup remove
update-rc.d: /etc/init.d/crio_control_loop_startup exists during rc.d purge (continuing)
Removing any system startup links for crio_control_loop_startup ...
/etc/rc0.d/K20crio_control_loop_startup
/etc/rc1.d/K20crio_control_loop_startup
/etc/rc2.d/S20crio_control_loop_startup
/etc/rc3.d/S20crio_control_loop_startup
/etc/rc4.d/S20crio_control_loop_startup
/etc/rc5.d/S20crio_control_loop_startup
/etc/rc6.d/K20crio_control_loop_startup
admin@NI-cRIO-9047:~#
```

# Demo: Full Application

Power cycle the cRIO

Wait roughly 30 seconds

Connect to the webpage in Chrome

- IPAddress:8181

Start the Python gRPC Client:

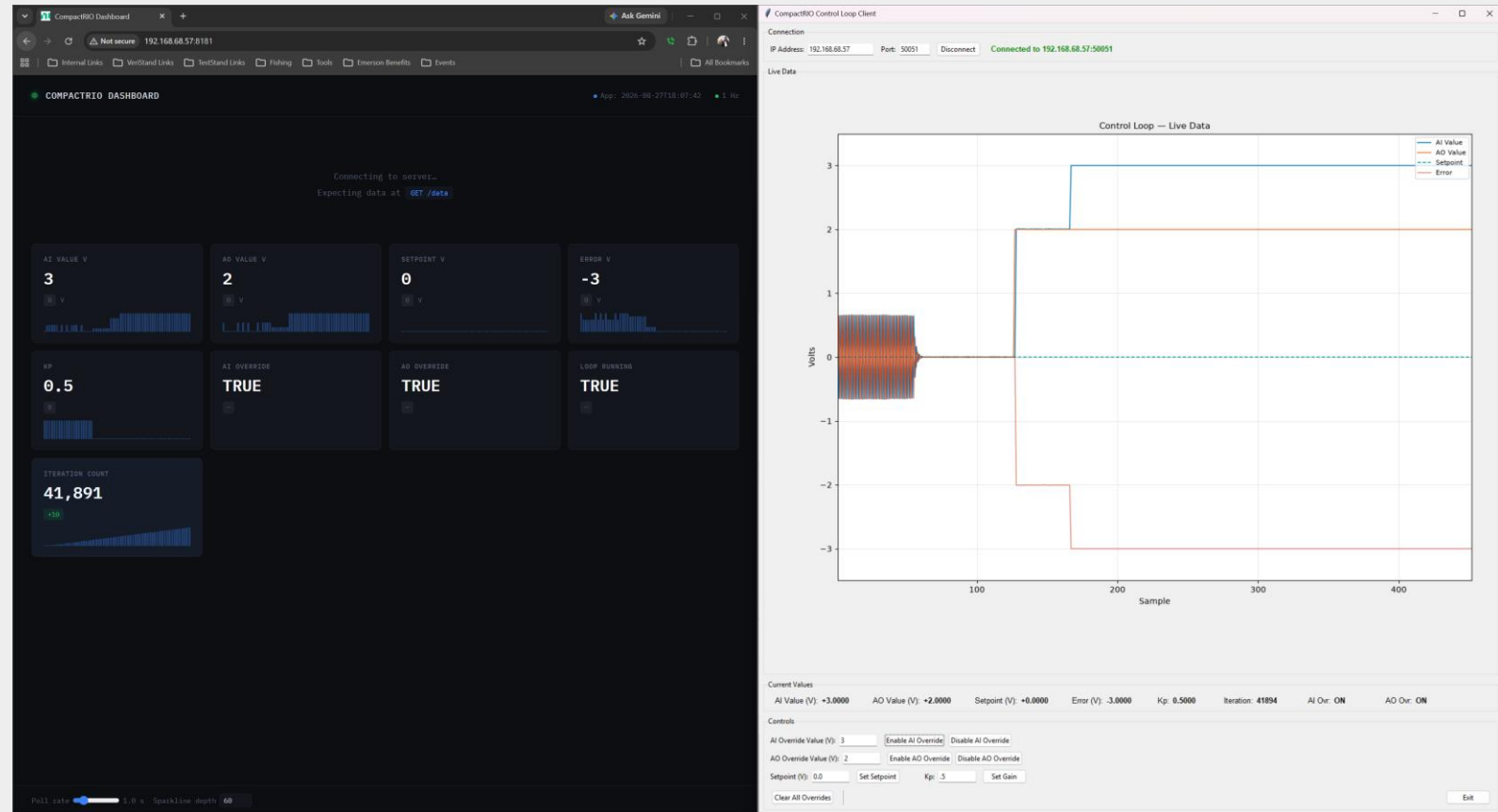
- control\_loop\_gui.py

Confirm pid file on cRIO:

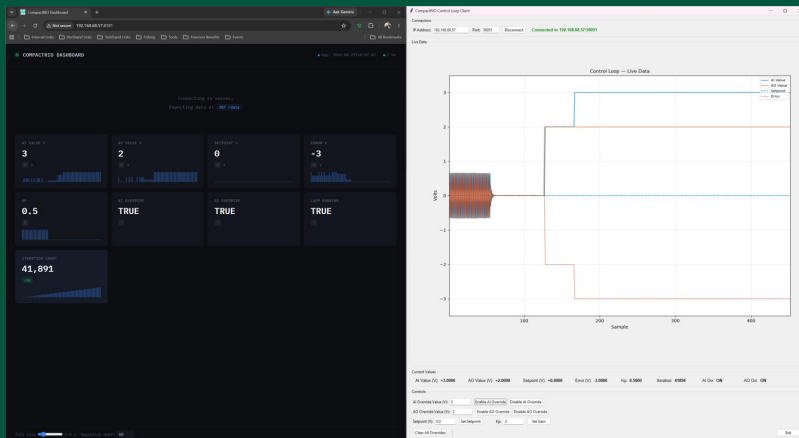
- /var/run/crio\_control\_loop\_startup.pid

Confirm log file on cRIO:

- /var/log/crio\_control\_loop\_startup.log



# DEMO: Full Application



cRIO uses SysV

Wait for drivers to initialize

gRPC is amazing...

Vibe coding is amazing...

**And Now...**

**Go and Vibe!!!**

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# Questions?

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We are here to help!

- Pull us in to design conversations
- Let us help de-risk projects
- Tell us what you want to learn about next